Retail Customer Retention Analytics – WALMART

## Project Overview:

Walmart needs a centralized Customer Retention Dashboard to transform its fragmented data into actionable insights, enabling analysis of customer churn, behavior across loyalty tiers, and the effectiveness of promotions and campaigns.

● Identify loyal vs. at-risk customers ● Measure the impact of loyalty tiers, promotions, and influencer-driven campaigns ● Guide region- and channel-specific retention strategies.

Project Objective: Develop an interactive Customer Retention Analytics Dashboard in Power BI using Walmart data that will: ● Consolidate customer demographics, purchase history, store/e-commerce performance, and membership data. ● Enable dynamic segmentation of high value, repeat, and churned customers. ● Providing actionable insights to improve retention, loyalty engagement, and regional strategies.

## Task 1: Data Modeling & Cleaning:

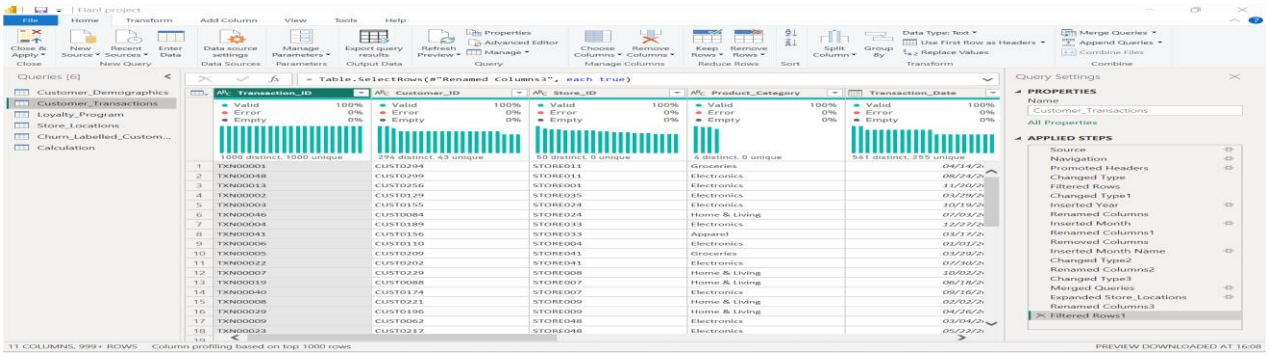
● Load and transform datasets in Power Query.

● Handle duplicates, missing values, and ensure correct data types.

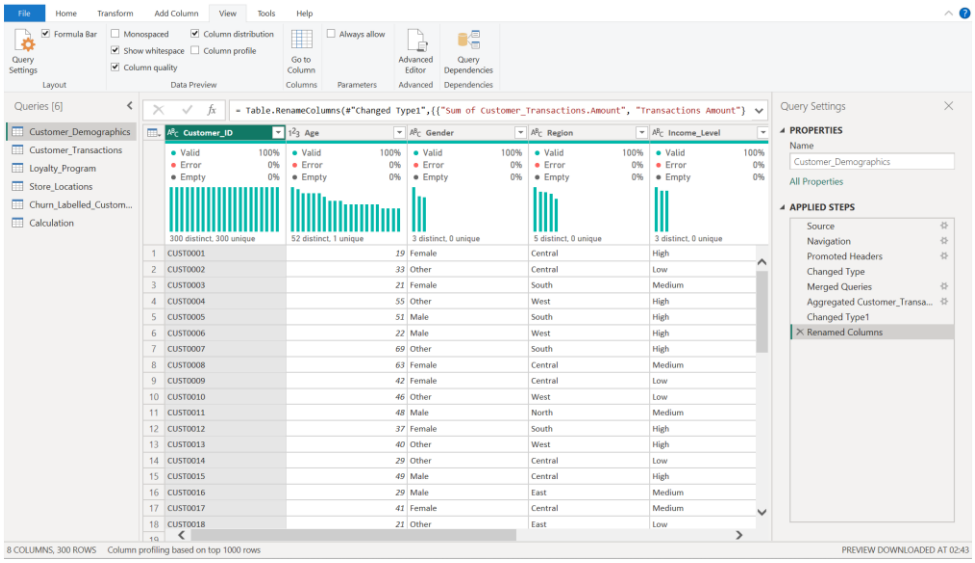
● Create calculated columns:

○ Membership Duration = Today – Membership Since.

```
Membership_Duration = (TODAY()-
Customer_Demographics[Membership_Since].[Date])/365
```



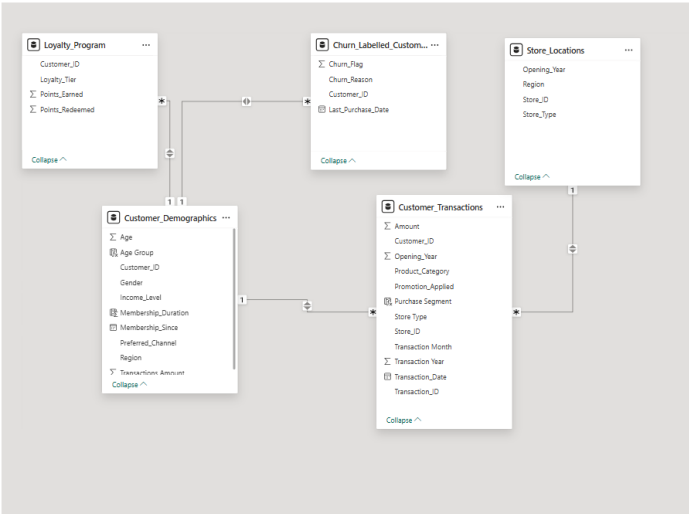
○ Extract Transaction Year, Transaction Month.




● Create a basic Data Model view

○ One-to-Many: Customer Demographics → Transactions, Loyalty Program, Churn Labelled Customers

○ Many-to-One: Transactions → Store Locations.



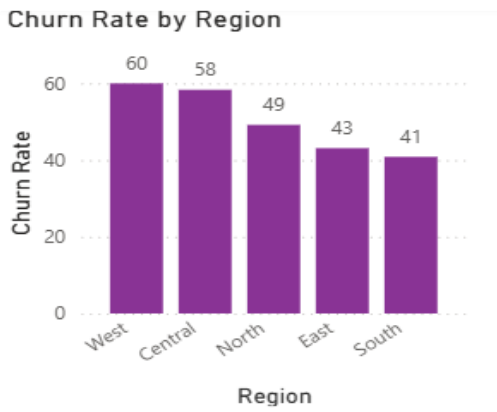
Task 2: Churn & Retention Metrics.

● Create Churn Rate KPI = (Churned Customers / Total Customers) * 100

```
Churn Rate = DIVIDE([Churned Customers],[Total customer])*100
```

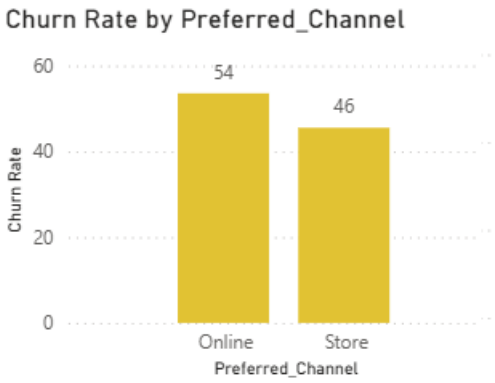
● Visualize churn rate by:

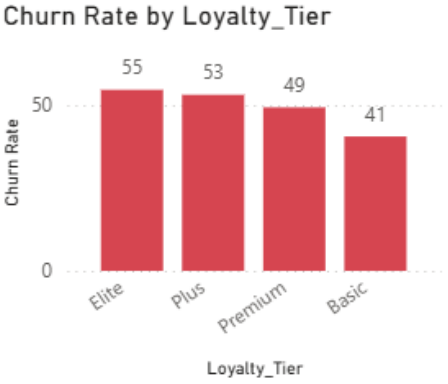
○ Region ○ Income Group ○ Channel (Store/Online) ○ Loyalty Tier.

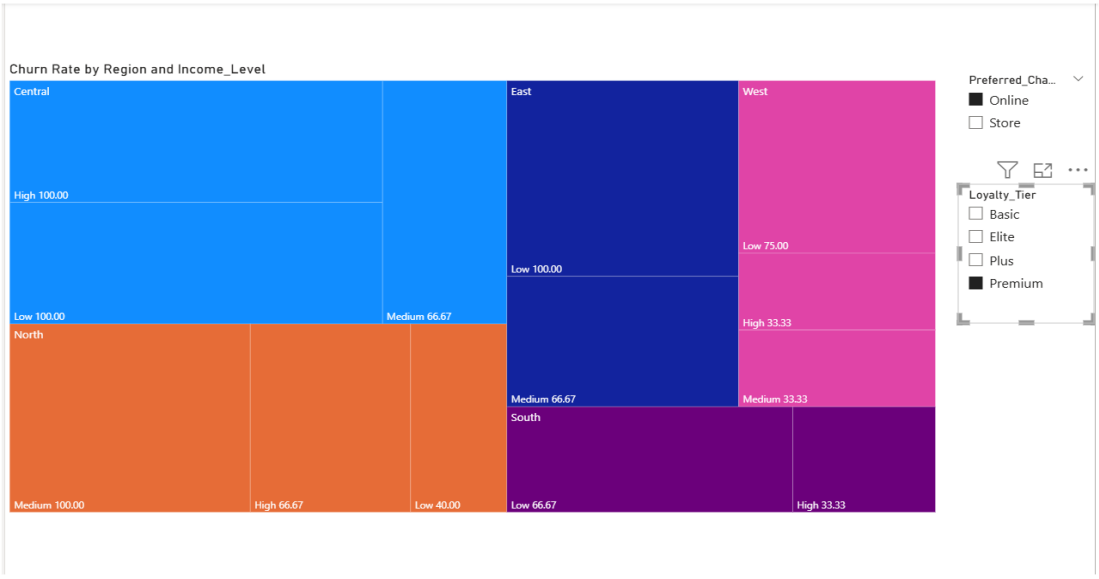




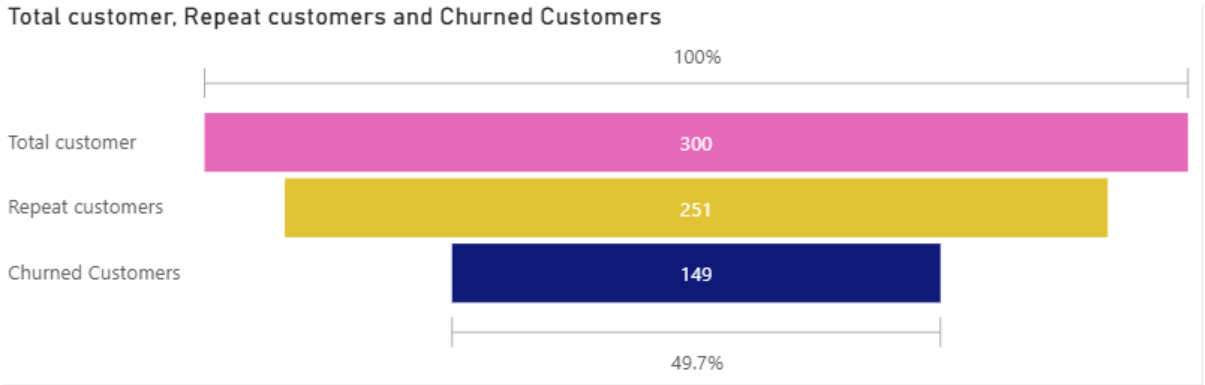







● Funnel Chart: Total Customers → Repeat Customers → Churned



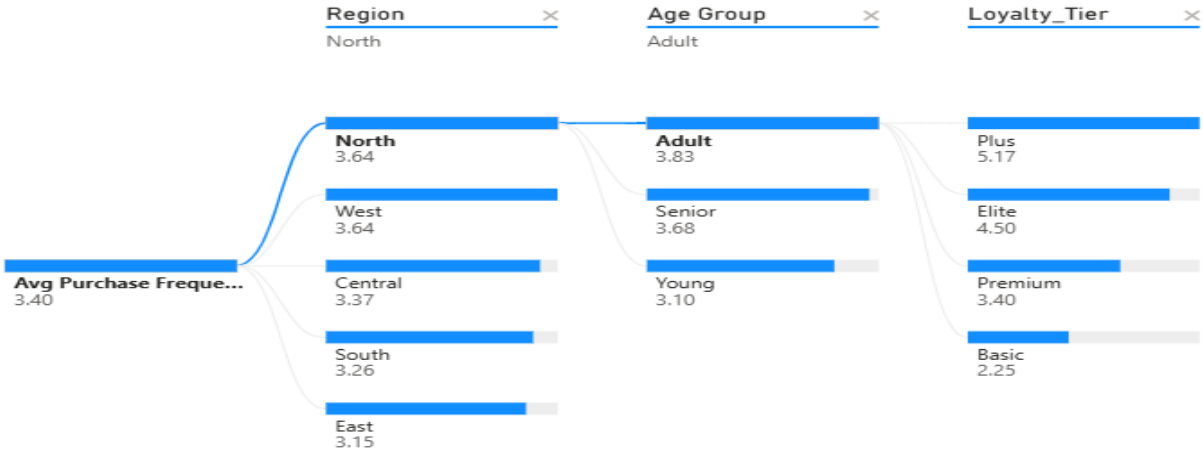

Task 3: Repeat Purchase Analysis.

● Segment customers:

○ Low-Tier: 0–3 purchases ○ Mid-Tier: 4–8 purchases ○ High-Tier: 9+ purchases.

```
Segment customers = SWITCH(TRUE(),[Purchase_Count]>=0 && [Purchase_Count]<=3, "Low-Tier (0-3)",[Purchase_Count]>=4 && [Purchase_Count]<=8, "Mid-Tier (4-8)",[Purchase_Count]>= "High-Tier (9+)")
```

● Compare avg. purchase frequency by Region, Age Group, Loyalty Tier.




● Identify most purchased product categories by loyal customers.

| |
|---|
| Loyal customers primarily purchase Groceries and Home & Living categories, indicating strong preference for essential and repeat-need products. These categories drive higher purchase frequency and sustained engagement among high-tier customers. Focusing loyalty rewards and personalized offers on these categories can further strengthen retention and |

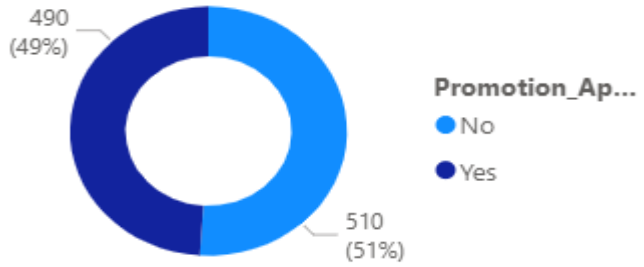
Task 4: Promotion & Loyalty Impact.

● % of transactions with promotion applied.

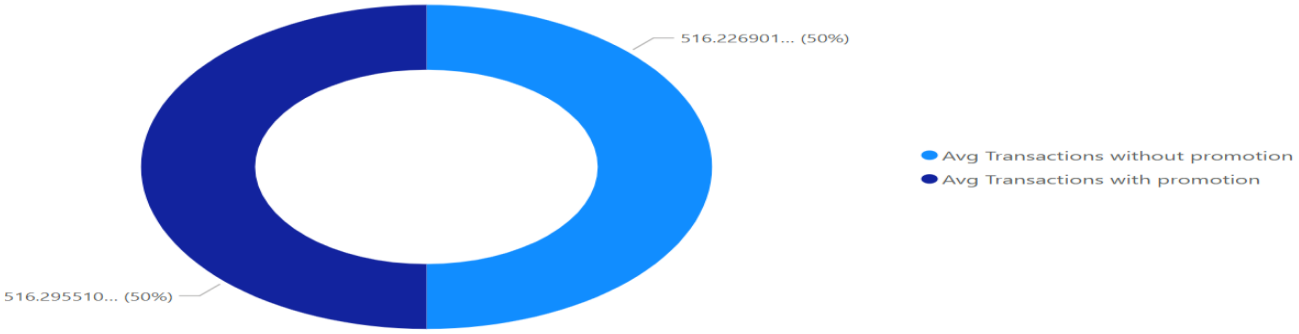
| |
|---|
| Avg Transactions with promotion = CALCULATE(AVERAGE(Customer_Transactions[Amount]),Customer_Transactions[Promotion_Applied]="Yes") |

| |
|---|
| Promotion Usage Rate with promotion applied (%) = ([promotion applied count]/COUNT(Customer_Transactions[Transaction_ID])*100) |

## % of Transactions With Promotion Applied



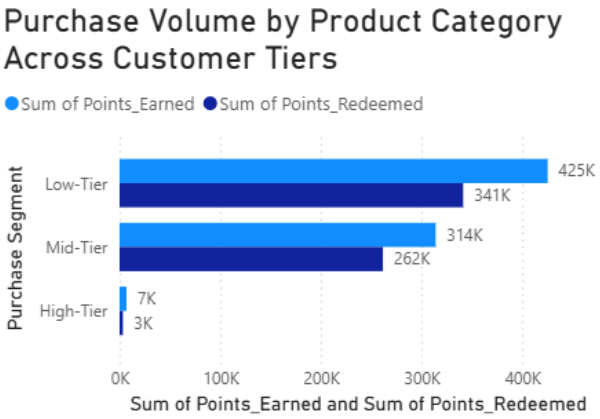

| Avg Transactions without promotion | Avg Transactions with promotion |
|---|---|
| 516.23 | 516.30 |



● Churn rate across loyalty tiers.




● Points Earned vs Redeemed by Tier (clustered column chart)

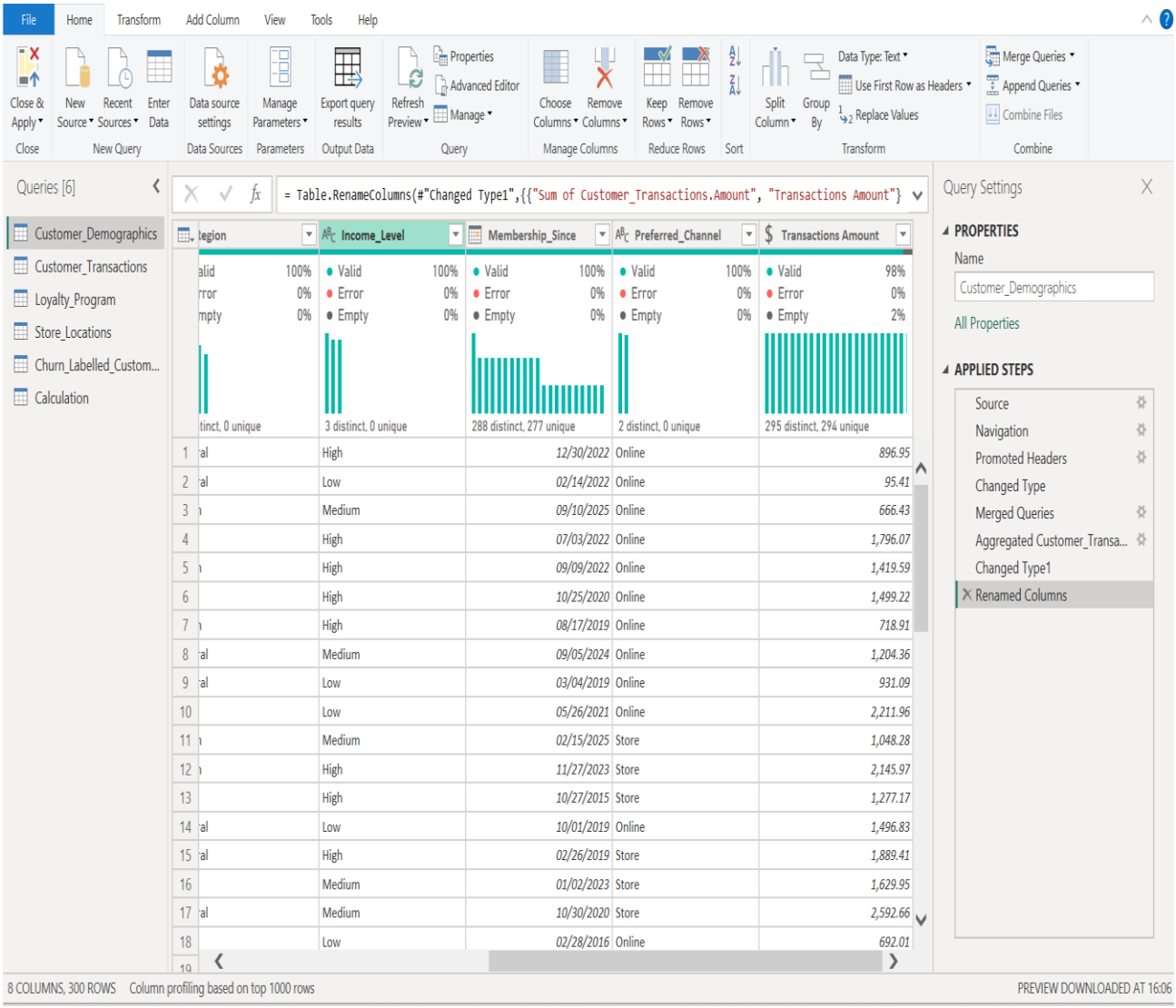



● Recommendations to improve redemption & retention.

| |
|---|
| Promotions are widely used but do not increase average purchase value, so Walmart should move from blanket discounts to targeted, loyalty-driven offers focused on high-frequency categories like Groceries. Simplifying reward redemption and applying tier- and store-specific incentives can improve engagement, increase redemption rates, and reduce churn across customer segments. |

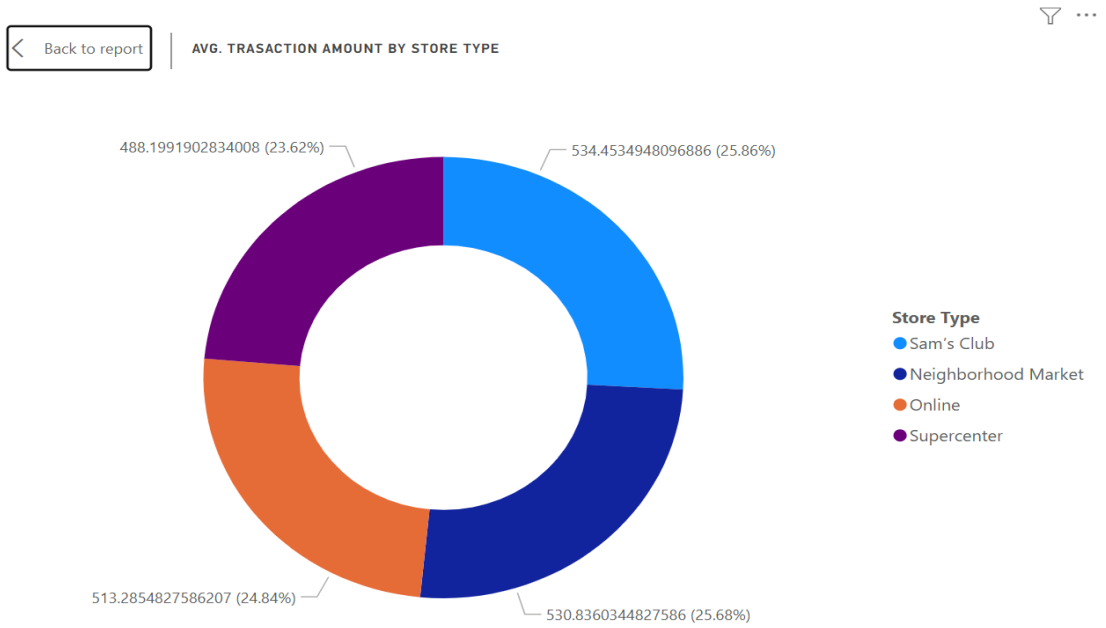
Task 5: Store & Channel Performance vs Retention.

● Merge store data with transactions.

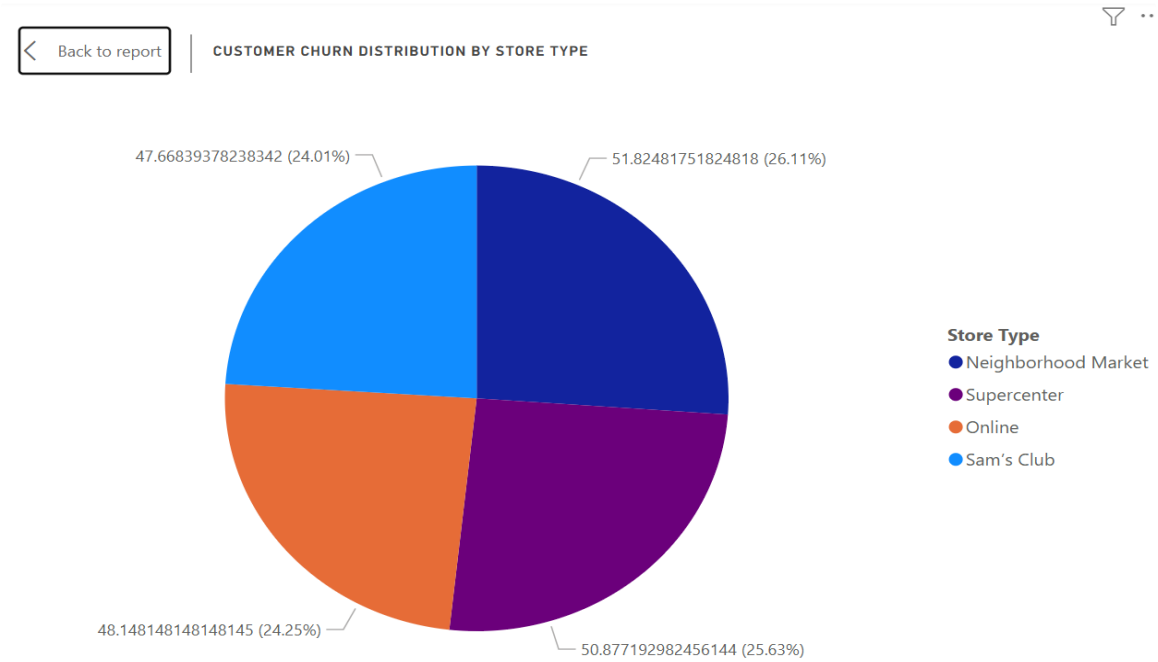



● Visualize:

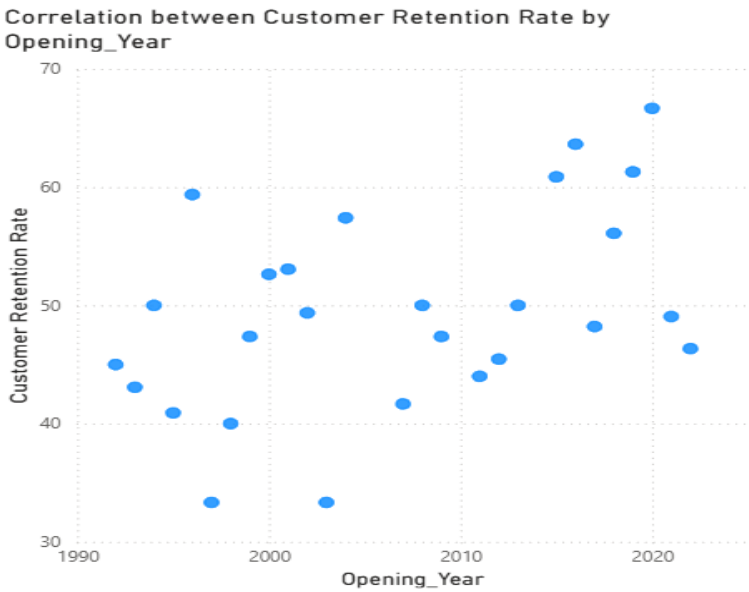
○ Avg. transaction amount by Store Type.




○ Churn rate by store type.




○ Correlation between store opening year & retention.




Task 6: Customer Lifetime Value (CLV) Analysis.

● CLV = Total Amount Spent / Membership Duration (Years).

```
Membership_Duration = (TODAY()-
Customer_Demographics[Membership_Since
].[Date])/365
```

```
Total Transaction Amount =
SUM(Customer_Transactions[Amount])
```

```
CLV =
DIVIDE(SUM(Customer_Transactions[Amount]),AVERAGE(Customer_
Demographics[Membership_Duration]),0)
```

● Segment customers into Low, High CLV

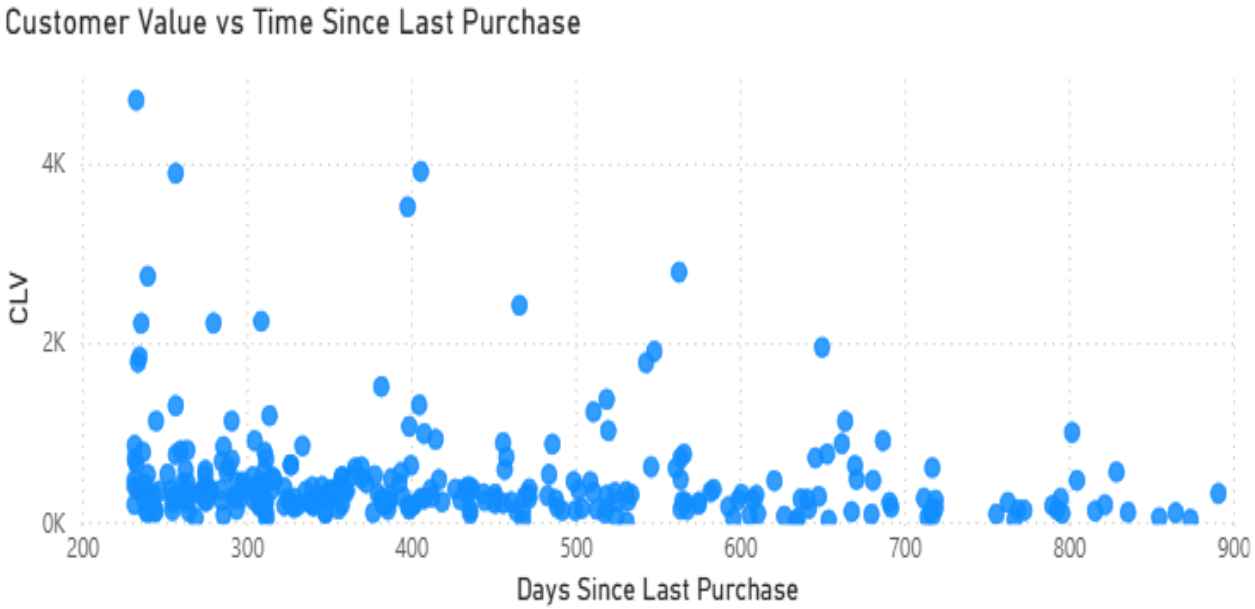
○ Above Average CLV as High ○ Below Average CLV value- Low

```
CLV Segment = IF('Calculation'[CLV] > CALCULATE([Average__CLV],
ALL('Customer_Demographics')),"High", IF('Calculation'[CLV] <
CALCULATE([Average__CLV], ALL('Customer_Demographics')),"Low"))
```

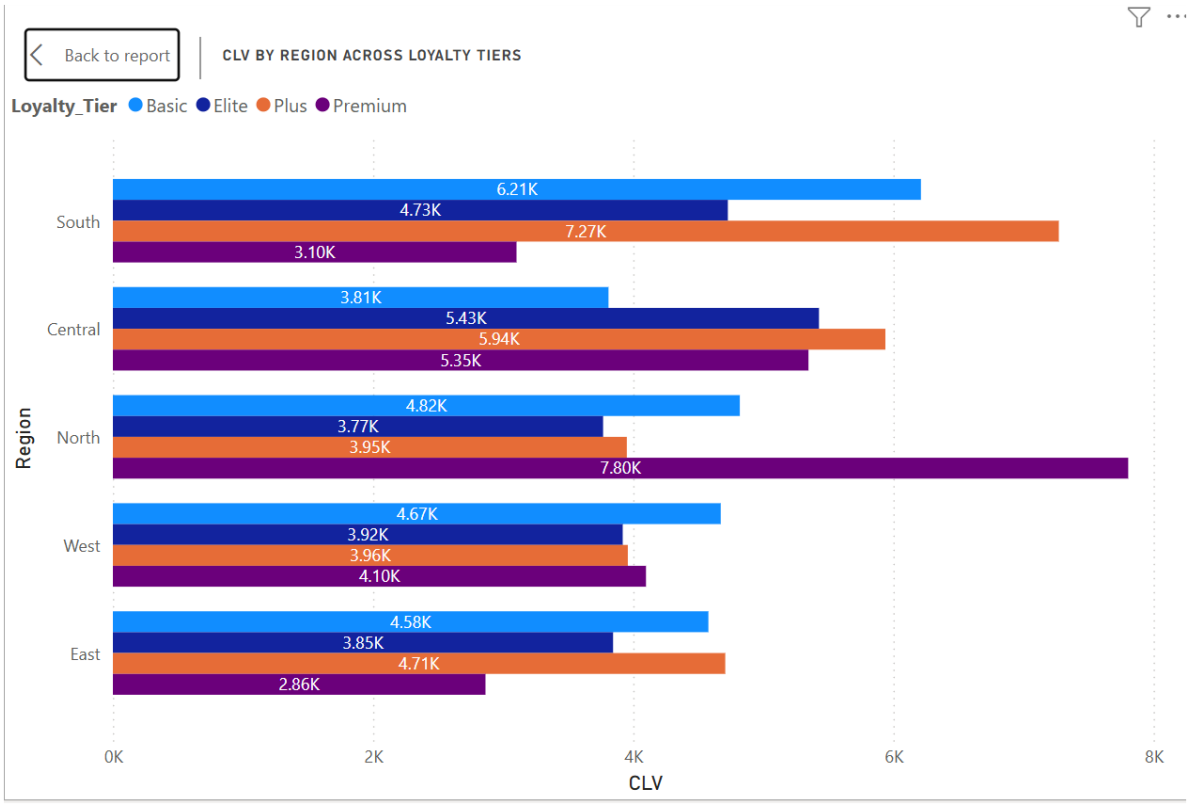
● Visualize:

○ CLV vs Days Since Last Purchase.

```
Days Since Last Purchase =
DATEDIFF(MAX(Customer_Transactions[Transaction_Date] ),T
ODAY(), DAY)
```



CLV by Loyalty Tier & Region.



Task 7: Final Dashboard & Executive Summary:

● Multi-page Power BI Report:

○ Page 1: KPIs (Churn, CLV, Repeat Rate).



```
Total customer = DISTINCTCOUNT(Customer_Demographics[Customer_ID])

Total Transaction = COUNT(Customer_Transactions[Transaction_ID])

Total Transaction Amount = SUM(Customer_Transactions[Amount])

Churned Customers =
CALCULATE(COUNT(Churn_Labelled_Customers[Customer_ID]),Churn_Labelled_Customers[Churn_Flag]=1)

Churn Rate = DIVIDE([Churned Customers],[Total customer])*100

Repeat customers =
CALCULATE(DISTINCTCOUNT(Customer_Transactions[Customer_ID]),FILTER(VALUES(Customer_Transactions[Customer_ID]),CALCULATE(COUNTROWS(Customer_Transactions))>1))

Repeat Rate = DIVIDE([Repeat customers],[Total customer])*100

CLV =
DIVIDE(SUM(Customer_Transactions[Amount]),AVERAGE(Customer_Demographics[Membership_Duration]),0)

Average__CLV = AVERAGEX(VALUES(Customer_Demographics[Customer_ID]),[CLV])

Purchase trasaction = DISTINCTCOUNT(Customer_Transactions[Transaction_ID])

Purchase_Count =
CALCULATE(COUNTROWS(Customer_Transactions),ALLEXCEPT(Customer_Transactions,Customer_Transactions[Customer_ID]))
```

○ Page 2: Loyalty & Promotion Impact.




Loyalty Redemption Rate = DIVIDE(SUM(Loyalty_Program[Points_Redeemed]),SUM(Loyalty_Program[Points_Earned]),0)

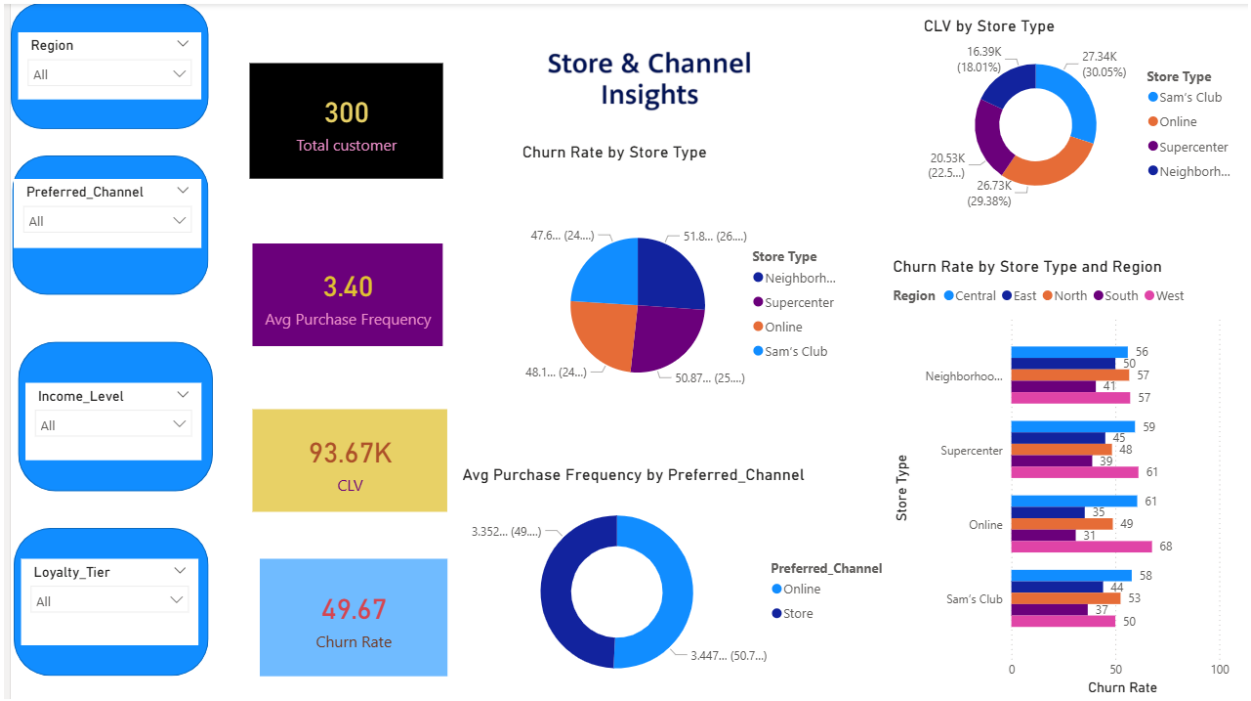
Promotion applied count = CALCULATE(COUNT(Customer_Transactions[Promotion_Applied]),Customer_Transactions[Promotion_Applied]="Yes")

Promotion Usage Rate with promotion applied (%) = ([promotion applied count]/COUNT(Customer_Transactions[Transaction_ID])*100)

Avg Transactions with promotion = CALCULATE(AVERAGE(Customer_Transactions[Amount]),Customer_Transactions[Promotion_Applied]="Yes")

Avg Transactions without promotion = CALCULATE(AVERAGE(Customer_Transactions[Amount]),Customer_Transactions[Promotion_Applied]="NO")

○ Page 3: Store/Channel Insights



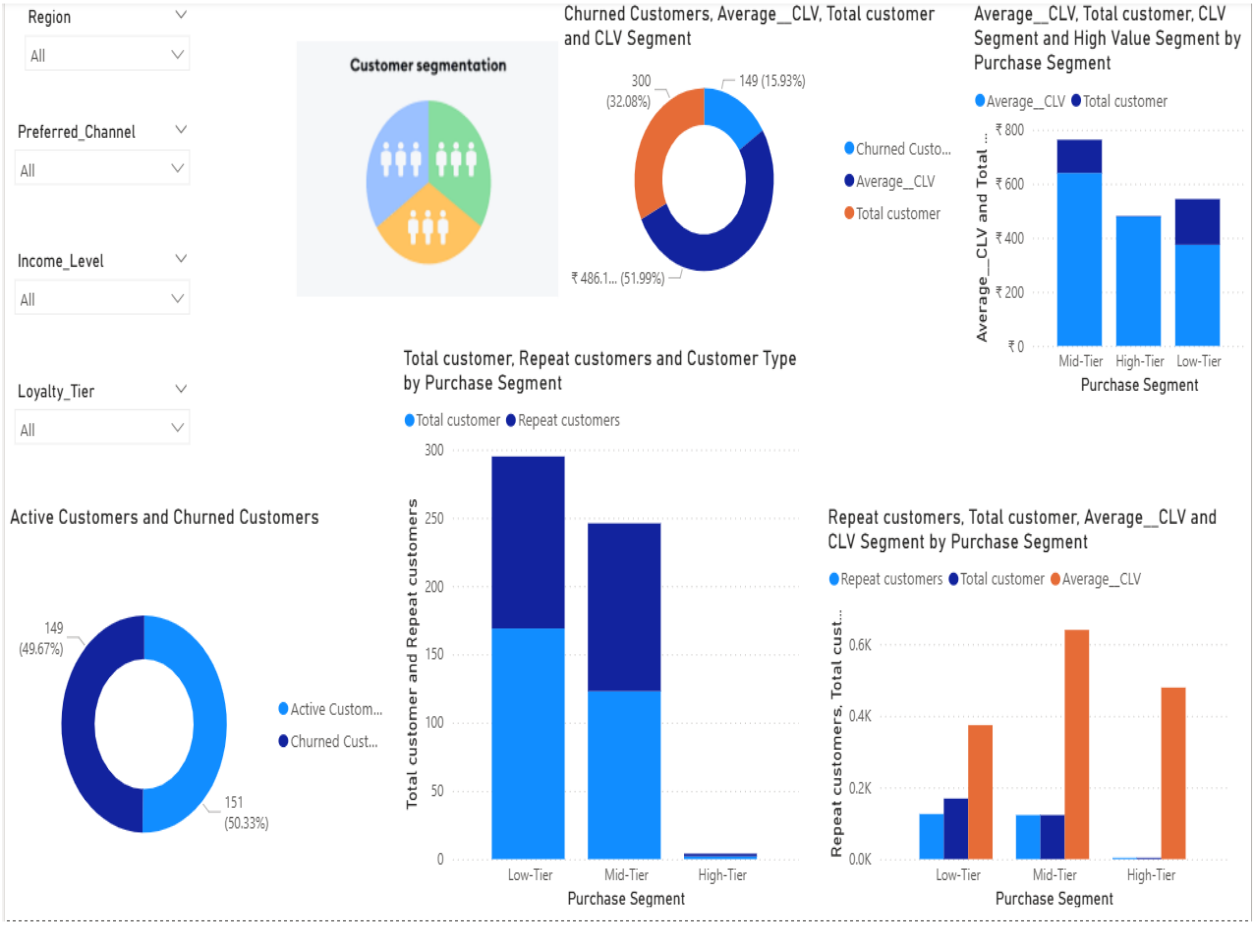

Avg Purchase Frequency = AVERAGEX(KEEPFILTERS(VALUES('Customer_Demographics'[Customer_ID])),[Purchase trasaction])

Customer Type = IF([Purchase_Count]>1,"Repeat","One-time")

○ Page 4: Segmentation (Churned, Repeat, High-Value)

● Slicers: Region, Channel, Income, Loyalty Tier.

```
Purchase Segment =
SWITCH(TRUE(),[Purchase_Count]<=3,"Low-
Tier",[Purchase_Count]<=8,"Mid-Tier","High-Tier")
```



● Top 3 recommendations for Walmart:

○ Which customers to prioritize for retention?

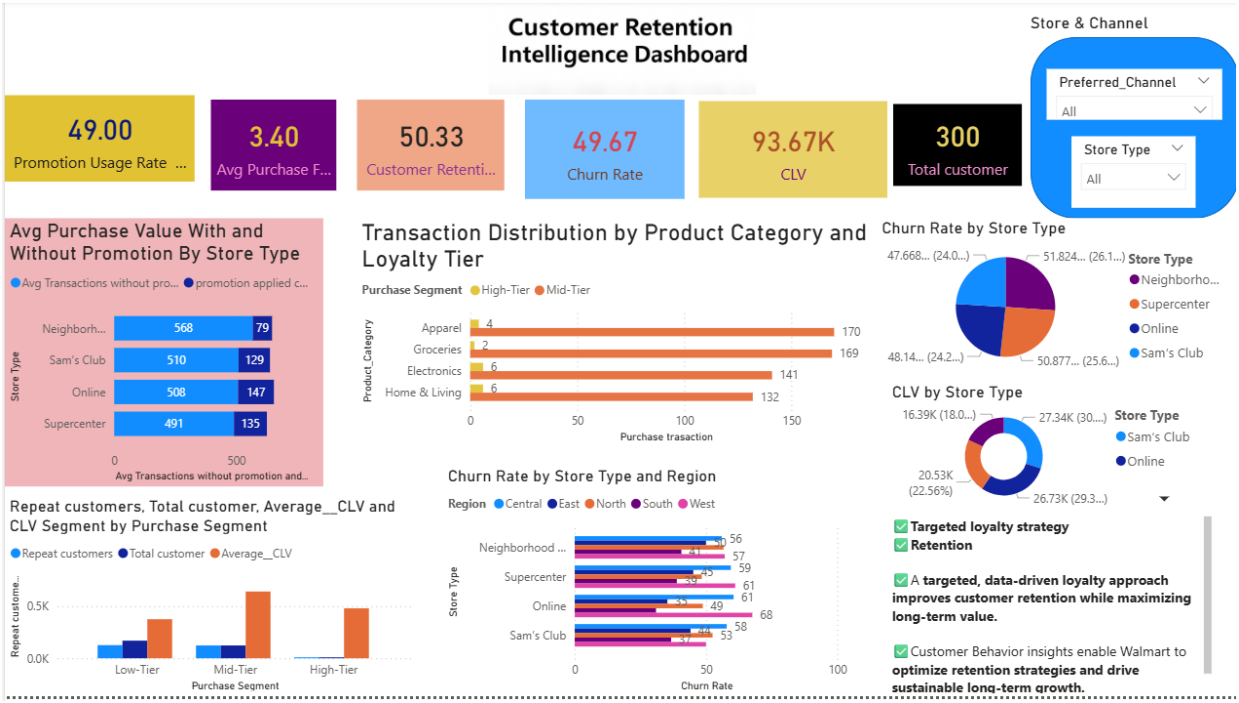
Walmart should prioritize **Repeat and High CLV customers**, especially those who are active but show signs of churn, as they contribute significantly to total customer value and long-term revenue.

○ Which channels are underperforming?

**Physical store channels** show relatively higher churn and lower purchase frequency compared to online channels, indicating a need for improved Instore engagement and targeted retention initiatives.

○ How to strengthen loyalty program engagement?

Loyalty engagement can be improved by **simplifying reward redemption** and offering **tier- and store specific incentives**, particularly for customers who earn points frequently but redeem them less often.



Video Link:

https://drive.google.com/file/d/1D2rtBRTB1Ueh2enFV33ECirnY8AUOO6Q/view?usp=drivesdk

Power Bi Project Link:

https://drive.google.com/file/d/1dfyCmV6ojkD6O8royyNHFD5XAjxWMGW3/view?usp=drivesdk